6/6

QUESTION 1: [6 POINTS]

Multiple choice questions. Select the correct answer.

A. [1 POINT] Which access modifier allows access only inside the same class?

a)	public.	b)	default.
c)	protected.	d)	private.

B. [1 POINT] What does it mean that a String in Java is immutable?

a)	Its value can be changed at any time.
b)	It cannot be modified after it is created.
c)	It can store only numbers.
d)	It is automatically deleted after use.

C. [1 POINT] What is the main difference between primitive and reference variables?

a)	Primitive stores objects, reference stores values.	b)	Both store memory addresses.
c)	Primitive stores values, reference stores memory addresses.	d)	Both store objects.

D. [1 POINT] What is the main goal of encapsulation?

a)	To speed up the program.	b)	To allow multiple inheritance.
c)	To hide and protect data.	d)	To avoid using objects.

E. [1 POINT] Which keyword is used to refer to the current object in Java?

a)	this.	b)	new.
c)	static.	d)	void.

F. [1 POINT] In aggregation, what is the relationship between objects?

a)	One object cannot exist without the other.	b)	One class inherits another.
c)	One object is part of another, but both can exist independently.	d)	Objects share the same memory.

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3.25 / 5.5

[5.5 POINTS]

QUESTION 3:

Write a **static method** named **countChar** that takes two parameters: a String named **sentence** and a char named **ch**. The method should count how many times the specified character appears in the sentence and return the count as an integer.

- The character comparison should be case-sensitive.
- The sentence may contain one or more characters, including spaces.

Sample 1:	Sample 2:
sentence: "Java is fun and Java is powerful" **ch:** 'a' **Output:** 5	**sentence:** "Hello world" **ch:** 'o' **Output:** 2

1.25

```
public static countChar(String sentence, char ch) {
static int count = 0;
    int stop = sentence.length();
    int n = 0;
    for(int i = 0; i < stop; i++)
        sentence.indexOf(ch) != -1
        count++;
    if(sentence.indexOf(ch, n++) != -1)
        count++;
}
}
```

B. [2 points] Evaluate the following Java codes and show the exact output:

```
public class TestString {
  public static void main(String[] args) {

    String msg = "Java is Powerful And Amazing";

    int p1 = msg.indexOf('P');
    int p2 = msg.lastIndexOf('a', msg.length() - 5);
    int p3 = msg.indexOf('a', p1 + 5);

    String sub1 = msg.substring(p1, p3);

    System.out.println(p1);
    System.out.println(p2);
    System.out.println(p3);
    System.out.println(sub1);
  }
}
```

8
23
1
power

Java is powerful And Amazing

length = 28

3.75 / 5.5

[5.5 POINTS]

QUESTION 2.

A. [3.5 points] Analyze each line of the following code and write in the table below whether the specified lines have an error or not. If the line has an error, then explain why. Assume that the two classes (Main and Laptop) are in the same package;

TestLaptop.java

```
public class TestLaptop {                                   //line1
  public static void main(String[] args) {                  //line2
    Laptop L1 = new Laptop();                               //line3
    Laptop L2 = new Laptop("Dell", 900.0);                  //line4
    System.out.println(L1.brand);                           //line5
    boolean check = Laptop.isAffordable(800);               //line6
    System.out.println(check);                              //line7
}}
```

Laptop.java

```
public class Laptop {                                       //line8
  private String brand;                                     //line9
  private double price;                                     //line10
  private int ram;                                          //line11
  public Laptop() {                                         //line12
    this.brand = "HP";                                      //line13
    price = 750.0;                                          //line14
  }

  public static Laptop(String brand, double price, int ram) {   //line15
    this.brand = brand;                                     //line16
    this.price = price;                                     //line17
    this.ram = ram;                                         //line18
  }
  public static boolean isAffordable(double maxPrice) {     //line19
    return(price <= maxPrice;                               //line20
}}
```

2.5 / 3.5

Line	Correct / Incorrect	Explanation why if incorrect
Line 15	Incorrect	A static method/constructor can't call or modify non static variables.
Line 20	Correct	
Line 3	Correct	
Line 5	Incorrect	brand is a private variable which can't be accessed without using a method like setBrand.

7/13

QUESTION 4: [13 POINTS]

Design an **Insurance Policy Management System** by creating two classes: Policy and Customer. The system should allow each customer to be associated with a policy, track their personal details and premium amount, and display complete information including policy data.
A **Customer owns a Policy**, meaning the Policy object is created and stored inside the Customer object and does not exist independently (tightly bound).

Suppose you have the following **Policy** class. Read it carefully and then complete the tasks below:

```
public class Policy {
    private String policyNumber;
    private String policyType;
    private double coverageAmount;

    public Policy(String policyNumber, String policyType, double
coverageAmount) {
        this.policyNumber = policyNumber;
        this.policyType = policyType;
        this.coverageAmount = coverageAmount;
    }

    public String getPolicyNumber() {
        return policyNumber;
    }

    public String getPolicyType() {
        return policyType;
    }

    public double getCoverageAmount() {
        return coverageAmount;
    }

    @Override
    public String toString() {
        return "[" + policyNumber + ", " + policyType + ", " +
coverageAmount + "]";
    }
}
```

Create a **Customer** class that represents a customer and contains the following:

A. **[1.5 points]** Private instance variables:

- customerId of type String – represents the customer's unique ID.
- customerName of type String – represents the customer's name.
- monthlyPremium of type double – represents the monthly premium.
- policy of type Policy – represents the policy associated with the customer.

A

1.5
1.5

```
public class Customer{
private String customerId;
private String CustomerName;
private double monthlypremium;
Policy policy;
}
```

B. **[4.5 points]** The Customer class must include:

- A parameterized constructor that initializes the customer's details and associates a Policy object with the Customer. The constructor should create or accept a Policy object to establish the relationship between the customer and their policy.

B

3.5
4.5

```
public class Customer{
public Customer(String customerId, String customerName,
double monthlypremium, String policyNumber, String PolicyType,
double coverageAmount){
Policy policy =new Policy(String policyNumber, String
PolicyType, double coverageAmount);}
}
```


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C. **[4.5 points]** An updatePremium() method that Use the policy object to determine the policyType for updating monthlyPremium This method should not return any value and should directly update the monthlyPremium field using the following rules:

- If the policyType is "Health" → increase monthly premium by 15%
- If the policyType is "Car" → increase monthly premium by 10%
- If the policyType is "Home" → increase monthly premium by 8%

0/4.5

D. **[2.5 points]** A toString() method that returns a formatted string containing the customer's details, including policy information.

2/2.5

```
public String toString(){
String ans = "Costumer ID: " + custumerId + ",customer Name is:" +
CustomerName + ",monthly premium = " +mouthlypremium + ",Policy number="
+ policyNumber + "policyType=" + policyType +
```

حسبي الله.

Good Luck ☺